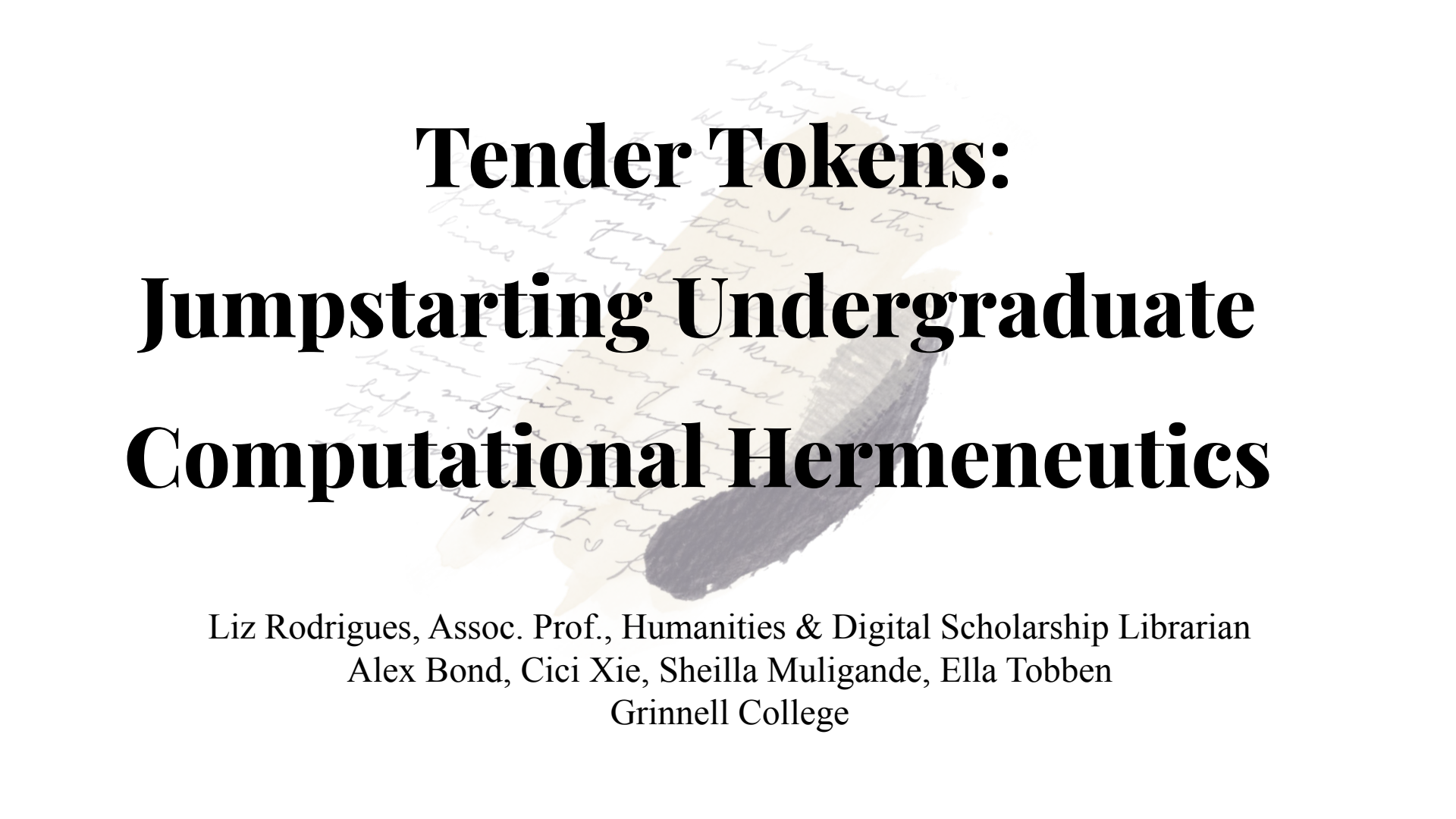




Tender Tokens: Jumpstarting Undergraduate Computational Hermeneutics

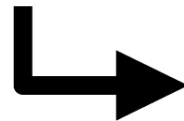
Liz Rodrigues, Assoc. Prof., Humanities & Digital Scholarship Librarian
Alex Bond, Cici Xie, Sheilla Muligande, Ella Tobben
Grinnell College



Overview & goals



<https://bit.ly/2026-tt-assignment>



TENDER
BUTTONS
GERTRUDE STEIN



Reading & Modeling *Tender Buttons*

A CARAFE, THAT IS A BLIND GLASS.¹

A kind in glass and a cousin, a spectacle and nothing strange a single hurt color and an arrangement in a system to pointing. All this and not ordinary, not unordered in not resembling. The difference is spreading.

“Bag-of-words” model

“a”, “kind”, “in”, “glass”, “and”, “a”, “cousin”,
“a”, “spectacle”, “and”, “nothing”, “strange”,
“a”, “single”, “hurt”, “color”, “and”, “an”
“arrangement”, “in”, “a”, “system”, “to”,
“pointing”, “all”, “this”, “and”, “not”,
“ordinary”, “not”, “unordered”, “in”, “not”,
“resembling”, “the”, “difference”, “is”,
“spreading”

Parts of speech model

DET N PREP N CONJ DET N DET N CONJ
ADV ADJ DET ADJ ADJ N CONJ DET N
PREP DET N PREP N

DET N CONJ ADV ADJ ADV ADJ PREP ADV
ADJ

DET N V V

¹ Gertrude Stein, *Tender Buttons* (1914; Project Gutenberg, March 17 2005),
“Objects”, <https://www.gutenberg.org/files/15396/15396-h/15396-h.htm>.

Beyond Counting Words:

What *Portraits and Repetition*¹ Taught Us

Our First Assumption:

Repetition



same word



same meaning

Stein's argument:

Repetition



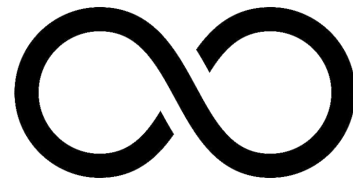
new context



new associations



new meaning



Repetition is not sameness.



¹ Gertrude Stein, "Portraits and Repetition," in Stein (New York: Penguin Books, 2001), 289–336.

LLM Experiment: Generating Stein-like Poetry

Project goal:

Test whether GPT-2 could generate a poem inspired by Gertrude Stein's *Tender Buttons*.

Why Stein?

She uses repetition, ordinary objects, colors, movement, and unusual syntax in a very intentional, but seemingly reproducible way.

Our key changes:

We changed the prompt to sound more Stein-like, then tested settings like `top_k`, temperature, and repetition blocking.

Our main question:

Can an LLM generate something Stein-like, or only imitate the surface patterns?

```
output_topk = model.generate(  
    input_ids,  
    max_length=max_length,  
    do_sample=True,  
    top_k=50,  
    pad_token_id=tokenizer.eos_token_id  
)  
print(tokenizer.decode(output_topk[0], skip_special_tokens=True))
```

```
output_topk = model.generate(input_ids, max_length=max_length,  
do_sample=True,  
  
| temperature=2.7, top_k=30, top_p=.97)  
print(tokenizer.decode(output_topk[0]))
```

```
# tokenize input prompt to sequence of ids  
prompt = "An orange."  
inputs = tokenizer(prompt, return_tensors="pt")  
  
# generate output as a sequence of token ids  
output = model.generate(  
    **inputs,  
    max_length=100,  
    num_return_sequences=1,  
    temperature=3.0,  
    top_k=100,  
    top_p=0.95,  
    repetition_penalty=0.95,  
    do_sample=True,  
    pad_token_id=tokenizer.eos_token_id,
```

```
for k in [5, 20, 50, 100]:  
    torch.manual_seed(8)  
    output = model.generate(  
        input_ids,  
        max_new_tokens=max_new_tokens,  
        do_sample=True,  
        top_k=k,  
        temperature=0.9,  
        pad_token_id=tokenizer.eos_token_id  
    )
```

Findings

Stein's unique language

Creativity vs temperature

Emphasis vs intention → is it repetition?

An Orange

An orange. Green in size - something for example something and things you bought. - it for the look if possible it made very pretty size it used in colour look! You used the very colour if. - usually and the size! You colour something made the things something used - usually pink because pretty Green thing made very size green for for. You for green pink make the and Green things. - green You in and green And pink were it the! -- green for and things thing thing

One of our "best attempts"; generated with GPT-2 by Alex Bond

Pedagogical takeaways and next steps

- Combining Steinian aesthetics and LLMs energizes the study of both
- More scaffolding necessary for classroom use
 - Introduce & produce an example of language modeling in class
 - Revise introductory LLM notebook to more explicitly guide experimentation
- Would probably take longer, perhaps a quarter of a semester

Thank you!

Email rodrigue8@grinnell.edu for further conversation

Assignment outline: <https://bit.ly/2026-tt-assignment>